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ARRANGEMENT STRUCTURE FOR COMMUNICATION DEVICE AND ELECTRONIC DEVICE INCLUDING THE SAME

Abstract

According to various embodiments, an electronic device includes a housing including a front surface plate; a rear surface plate facing toward the opposite direction of the front surface plate; and a side surface member surrounding a space between the front surface plate and the rear surface plate, the side surface member having a substantially rectangular shape when viewed above the front surface plate; a first PCB arranged in the space; a first wireless communication circuit; a substrate; a first antenna array protruding from the first side of the substrate toward the first portion; a second antenna array protruding from the second side of the substrate toward the second portion; and a second wireless communication circuit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of application Ser. No. 18/464,971, filed Sep. 11, 2023, which is a continuation of application Ser. No. 17/099,615, filed Nov. 16, 2020, now U.S. Pat. No. 11,757,174, which is a continuation of application Ser. No. 16/201,500, filed Nov. 27, 2018, now U.S. Pat. No. 10,840,583, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2017-0159219, filed on Nov. 27, 2017, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entirety.

BACKGROUND

1. Field

[0002] Various embodiments of the present disclosure relate to an arrangement structure for a communication device and an electronic device including the same.

2. Description of Related Art

[0003] With the development of wireless communication technology, electronic devices (for example, electronic devices for communication) are popularly used in our daily lives, and accordingly, use of contents is increasing exponentially. As use of contents is rapidly increasing, network capacities reach their limits. As low latency data communication is required, the introduction of next-generation wireless communication technology (for example, 5G communication) or high-speed wireless communication such as wireless gigabit alliance (WIGIG) (for example, 802.11AD) may be required.

[0004] The above information is presented as background information only to assist with an understanding of the present disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the present disclosure.

SUMMARY

[0005] Next-generation wireless communication technology may use a millimeter wave of substantially 25 GHz or higher, and a communication device corresponding thereto may include a plurality of antenna arrays arranged on one surface of a dielectric (for example, a substrate), and a communication circuit (for example, a radio frequency integrated circuit (RFIC)) arranged on the other surface and electrically connected with the antenna arrays. According to an embodiment, the communication device may form a beam in a predetermined direction through a plurality of antenna elements, and may radiate a phase-adjusted signal in a specified direction.

[0006] The above-described next-generation wireless communication device provided in a module

form may be substantially applied to a rear surface or a peripheral portion of an electronic device in order to overcome a beam coverage, and may be arranged to avoid interference by an existing communication device (for example, a 4th generation wireless communication device).

[0007] Various embodiments of the present disclosure provide an arrangement structure for a communication device and an electronic device including the same.

[0008] Various embodiments of the present disclosure provide an arrangement structure for a communication device, which has a plurality of communication devices having the same configuration and arranged in various positions of an electronic device without changing designs, and an electronic device including the same.

[0009] According to an embodiment of the present disclosure, an electronic device includes: a housing including: a front surface plate; a rear surface plate facing toward the opposite direction of the front surface plate; and a side surface member surrounding a space between the front surface plate and the rear surface plate, the side surface member having a substantially rectangular shape when viewed above the front surface plate, the side surface member including: a first portion having a first length and extending in a first direction; a second portion having a second length longer than the first length, and extending in a second direction perpendicular to the first direction; a third portion having the first length and extending in the first direction in parallel with the first portion; and a fourth portion having the second length and extending in the second direction in parallel with the second portion; a first PCB arranged in the space; a first wireless communication circuit which is mounted on the first PCB and is electrically connected with a first point of the first portion of the side surface member, and provides wireless communication ranging from 500 MHz to 6000 MHz; a substrate arranged close to a corner in the space in parallel with the rear surface plate, the substrate including: a first side extending along a part of the first portion, a second side extending along the second portion; a third side extending in parallel with the first side; and a fourth side extending in parallel with the second side; a first antenna array protruding from the first side of the substrate toward the first portion; a second antenna array protruding from the second side of the substrate toward the second portion; and a second wireless communication circuit attached to a first part on the third side closer to the fourth side than the second portion, or attached to a second part on the fourth side closer to the third side than the first portion, wherein the second wireless communication circuit is electrically connected to the first antenna array and the second antenna array, and provides wireless communication ranging from 20 GHz to 100 GHz.

[0010] According to various embodiments, an electronic device includes: a housing including: a front surface plate; a rear surface plate facing toward the opposite direction of the front surface plate; and a side surface member surrounding a space between the front surface plate and the rear surface plate, the side surface member having a substantially rectangular shape when viewed above the front surface plate, the side surface member including: a first portion having a first length and extending in a first direction; a second portion having a second length longer than the first length and extending in a second direction perpendicular to the first direction; a third portion having the first length and extending in the first direction in parallel with the first portion; and a fourth portion having the second length and extending in the second direction in parallel with the second portion; a first substrate arranged in the space; a first wireless communication circuit which is mounted on the first substrate and is electrically connected with a first point of the first portion of the side surface member, and provides wireless communication of a first frequency band; a second substrate arranged in the proximity of a corner between the first portion and the second portion in the space in parallel with the rear surface plate, the second substrate including: a first side extending along a part of the first portion, a second side extending along the second portion; a third side extending in parallel with the first side; and a fourth side extending in parallel with the second side; a first antenna array arranged to form a beam pattern in a direction from the first side of the second substrate toward the first portion; a second antenna array arranged to form a beam pattern in a direction from the second side of the second substrate toward the second portion; a third substrate

having a same configuration as that of the second substrate, and arranged in the proximity of a corner between the second portion and the third portion in the space in parallel with the rear surface plate, after being rotated by 90 degrees in the clockwise direction with reference to the arrangement state of the second substrate; a fourth substrate having a same configuration as that of the second substrate, and arranged in the proximity of a corner between the third portion and the fourth portion in the space in parallel with the rear surface plate, after being rotated by 90 degrees in the clockwise direction with reference to the arrangement state of the third substrate; a fifth substrate having a same configuration as that of the second substrate, and arranged in the proximity of a corner between the fourth portion and the first portion in the space in parallel with the rear surface plate, after being rotated by 90 degrees in the clockwise direction with reference to the arrangement state of the fourth substrate; and a second wireless communication circuit attached to a first part on the third side closer to the fourth side than the second portion, or attached to a second part on the fourth side closer to the third side than the first portion, wherein the second wireless communication circuit is electrically connected to the first antenna array and the second antenna array and provides wireless communication in a second frequency band.

[0011] The electronic device according to various embodiments of the present disclosure has a plurality of communication devices having the same configuration, and arranged on various positions of the electronic device without changing designs, such that an increase in the number of processes and an increase in costs by a design change of the communication device can be prevented, and radiation performance of the communication device can be enhanced.

[0012] Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document: the terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation; the term “or,” is inclusive, meaning and/or; the phrases “associated with” and “associated therewith,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, or the like; and the term “controller” means any device, system or part thereof that controls at least one operation, such a device may be implemented in hardware, firmware or software, or some combination of at least two of the same. It should be noted that the functionality associated with any particular controller may be centralized or distributed, whether locally or remotely.

[0013] Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

[0014] Definitions for certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many, if not most instances, such definitions apply to prior, as well as future uses of such defined words and phrases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0016] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various embodiments of the present disclosure;

[0017] FIGS. 2A and 2B are perspective views illustrating an electronic device according to various embodiments of the present disclosure;

[0018] FIG. 3 is a view illustrating an arrangement relationship of a communication device in an electronic device according to various embodiments of the present disclosure;

[0019] FIGS. 4A and 4B are perspective views illustrating a communication device according to various embodiments of the present disclosure;

[0020] FIG. 5 is a view illustrating an arrangement of a communication device according to various embodiments of the present disclosure;

[0021] FIGS. 6A and 6B are perspective views illustrating a communication device according to various embodiments of the present disclosure;

[0022] FIGS. 7A and 7B are views illustrating an arrangement of a communication device according to various embodiments of the present disclosure;

[0023] FIGS. 8A-8C are views illustrating an arrangement of a communication device according to various embodiments of the present disclosure;

[0024] FIG. 9 is a view illustrating an arrangement structure of communication devices of an electronic device according to various embodiments of the present disclosure;

[0025] FIGS. 10A-10C are views illustrating a configuration of a housing of an electronic device according to various embodiments of the present disclosure;

[0026] FIG. 11A is a view illustrating an arrangement relationship between an existing housing and a communication device, and a beam pattern direction of the communication device; and

[0027] FIG. 11B is a view illustrating an arrangement relationship between an existing housing and a communication device according to various embodiments of the present disclosure, and a beam pattern direction of the communication device.

DETAILED DESCRIPTION

[0028] FIGS. 1 through 11B, discussed below, and the various embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged system or device.

[0029] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various embodiments of the present disclosure.

[0030] Referring to FIG. 1, the electronic device **101** in the network environment may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, a memory **130**, an input device **150**, a sound output device **155**, a display device **160**, an audio module **170**, a sensor module **176**, an interface **177**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module **196**, and an antenna module **197**. In some embodiments, at least one (e.g., the display device **160** or the

camera module **180**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components may be implemented as single integrated circuitry. For example, the sensor module **176** (e.g., a fingerprint sensor, an iris sensor, or an illuminance sensor) may be implemented as embedded in the display device **160** (e.g., a display).

[0031] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. The processor **120** may load a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), and an auxiliary processor **123** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from the main processor **121**. Additionally or alternatively, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as embedded in the main processor **121**.

[0032] In this case, the auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display device **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0033] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0034] The input device **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input device **150** may include, for example, a microphone, a mouse, or a keyboard.

[0035] The sound output device **155** may output sound signals to the outside of the electronic device **101**. The sound output device **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record, and the receiver may be used for incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0036] The display device **160** may visually provide information to the user of the electronic device **101**. The display device **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding device. According to an embodiment, the display device **160** may include touch circuitry or a pressure sensor adapted to measure the intensity of force incurred by a touch.

[0037] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input device **150**, or output the sound via the sound output device **155** or an external electronic device (e.g., the

electronic device **102**) (for example, a speaker or a headphone) wiredly or wirelessly coupled with the electronic device **101**.

[0038] The sensor module **176** may generate an electrical signal or data value corresponding to an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state external to the electronic device **101**. The sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0039] The interface **177** may support a specified protocol to be coupled with the external electronic device (e.g., the electronic device **102**) wiredly or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0040] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**), for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0041] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. The haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0042] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0043] The power management module **188** may manage power supplied to the electronic device **101**, and may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0044] The battery **189** may supply power to at least one component of the electronic device **101**. The battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0045] The communication module **190** may support establishing a wired or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a wired or wireless communication.

[0046] According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as BLUETOOTH™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a cellular network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules **190** may be implemented as a single chip, or may be implemented as multi chips separate from each other.

[0047] According to an embodiment, the wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, using user information stored in the subscriber identification module **196**.

[0048] The antenna module **197** may include one or more antennas to transmit or receive a signal or power to or from the outside. According to an embodiment, the communication module **190** (e.g., the wireless communication module **192**) may transmit or receive a signal to or from the external electronic device through an antenna appropriate for a communication scheme.

[0049] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0050] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** and **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices. According to an embodiment, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, or client-server computing technology may be used, for example.

[0051] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smart phone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the present disclosure, the electronic devices are not limited to those described above.

[0052] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, and/or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and/or B,” “at least one of “A, B, or C,” or “A, B, and/or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st,” “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly or via a third element.

[0053] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry.” A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0054] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a machine-readable storage medium (e.g.,

internal memory **136** or external memory **138**) that is readable by a machine (e.g., a computer). The machine may invoke an instruction stored in the storage medium, and may be operated according to the instruction invoked, and may include an electronic device (e.g., the electronic device **101**) according to disclosed embodiments. When the instruction is executed by the processor (e.g., the processor **120**), the processor may execute a function corresponding the instruction, with or without using one or more other components under the control of the processor. The instruction may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal, but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0055] According to an embodiment, a method according to various embodiments of the present disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed online via an application store (e.g., PLAY STORE™). If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0056] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. One or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0057] FIGS. 2A and 2B are perspective views illustrating an electronic device according to various embodiments of the present disclosure.

[0058] FIG. 2A is a front surface perspective view of the electronic device, and FIG. 2B is a rear surface perspective view of the electronic device.

[0059] The electronic device **200** of FIGS. 2A and 2B may be similar to the electronic device **101** of FIG. 1 at least in part, or may include other embodiments of the electronic device.

[0060] Referring to FIGS. 2A and 2B, the electronic device **200** may include a housing **210**. According to an embodiment, the housing **210** may be formed of a conductive member and/or a nonconductive member. According to an embodiment, the housing **210** may include a first surface **2001** (for example, a front surface or a top surface) facing toward a first direction (for example, a Z-axis direction), a second surface **2002** (for example, a rear surface or a bottom surface) arranged opposite the first surface **2001**, and a side surface **2003** arranged to surround at least part of the first surface **2001** and the second surface **2002**. According to an embodiment, the side surface **2003** may be coupled with a front surface plate **2011** (for example, a glass plate including various coating layers or a polymer plate) and a rear surface plate **211**, and may be formed by a side surface member **216** including metal and/or polymer. According to an embodiment, the rear surface plate **211** may be formed from coted or colored glass, ceramic, polymer, metal (for example, aluminum, stainless steel (STS), or magnesium), or a combination of at least two materials of the above-described materials.

[0061] According to various embodiments, the side surface **2003** may be coupled with the front surface plate **2011** and the rear surface plate **211**, and may be formed by the side surface member **216** (or a “side surface bezel structure”) including metal and/or polymer. According to an embodiment, the rear surface plate **211** and the side surface member **216** may be integrally formed with each other, and may include the same material (for example, a metallic material such as aluminum or magnesium). According to an embodiment, the housing **210** may include a first portion **2101** having a first length, a second portion **2102** extending perpendicular to the first portion **2101** and having a second length, a third portion **2103** extending from the second portion **2102** to have the first length in parallel with the first portion **2101**, and a fourth portion **2104** extending from the third portion **2103** to have the second length in parallel with the second portion **2102**. According to an embodiment, the first portion **2101** may have a unit conductive portion **2101** electrically isolated by a pair of nonconductive portions **223**, **224** spaced apart from each other by a predetermined distance. In addition, the third portion **2103** may also have a unit conductive portion **2103** electrically isolated by a pair of nonconductive portions **221**, **222** spaced apart from each other by a predetermined distance. However, this should not be considered as limiting. The first portion **2101** and the third portion **2103** formed as a unit conductive portion may be formed by one or more nonconductive portions. The conductive portions **2101**, **2103** electrically isolated may be electrically connected with a communication circuit (for example, a communication circuit **390** of FIG. **3**) arranged inside the electronic device **200**, and may be utilized as antennas operating in at least one resonant frequency band.

[0062] According to various embodiments, the electronic device **200** may include the front surface plate **2011** (for example, a window or glass plate) arranged on the first surface **2001**, and a display **201** (for example, a touch screen display) arranged to be exposed through at least one portion of the front surface plate **2011**. According to an embodiment, the display **201** may be coupled with a touch detection circuit, a pressure sensor for measuring an intensity (pressure) of a touch, and/or a pen detection sensor (for example, a digitizer) for detecting a stylus pen of a magnetic field scheme, or may be arranged adjacent thereto.

[0063] According to various embodiments, the electronic device **200** may include a receiver hole **202** for communication. According to an embodiment, the electronic device **200** may be controlled to use a speaker arranged therein and to talk with the other person through the receiver hole **202** for communication. According to an embodiment, the electronic device **200** may include a microphone hole **203**. According to an embodiment, the electronic device **200** may use at least one microphone arranged therein to detect a direction of a sound, and may receive an external sound or transmit a user's voice to the other person through the microphone hole **203**.

[0064] According to various embodiments, the electronic device **200** may include at least one key input device **217**. According to an embodiment, the key input device **217** may include at least one side key button **217** arranged on the side surface **2003** of the housing **210**. According to an embodiment, the at least one side key button **217** may include a volume control button, a wake-up button, or a specific function (for example, an artificial intelligence execution function or a rapid voice recognition execution mode entering function) performance button.

[0065] According to various embodiments, the electronic device **200** may include components which are exposed through the display **201**, or are arranged to perform functions through the front surface plate **2011**, but not to exposed, and performs various functions of the electronic device **200**. According to an embodiment, at least part of the components may be arranged to be in contact with an external environment from the inside of the electronic device through at least one portion of the front surface plate **2011** of a transparent material. According to an embodiment, the components may include at least one sensor module **204**. The sensor module **204** may include, for example, an illuminance sensor (for example, a light sensor), a proximity sensor (for example, a light sensor), an infrared sensor, an ultrasound sensor, a fingerprint recognition sensor, a face recognition sensor, an electromagnetic (EM) sensor, or an iris recognition sensor. According to an embodiment, the

components may include a first camera device **205**. According to an embodiment, the components may include an indicator **206** (for example, an LED device) for visually providing state information of the electronic device **200** to the user. According to an embodiment, the components may include a light source **214** (for example, an infrared LED) arranged at one side of the receiver hole **202**. According to an embodiment, the components may include an imaging sensor assembly **215** (for example, an iris camera) for detecting an iris image when light generated from the light source **214** is irradiated onto the vicinity of user's eyes. According to an embodiment, at least one of the components may be arranged to be exposed through at least one portion of the second surface **2002** (for example, a rear surface or a bottom surface) facing toward a direction (for example, the $-Z$ axis direction) opposite the first direction of the electronic device **200**.

[0066] According to various embodiments, the electronic device **200** may include an external speaker hole **207**. According to an embodiment, the electronic device **200** may use a speaker arranged therein, and may emit a sound through the external speaker hole **207**. According to an embodiment, the electronic device **200** may include a first connector hole **208** (for example, an interface connector port) for performing a data exchanging function with an external device, and for receiving external power and charging the electronic device **200**. According to an embodiment, the electronic device **200** may include a second connector hole **209** (for example, an ear jack assembly) for receiving an ear jack of an external device.

[0067] According to various embodiments, the electronic device **200** may include the rear surface plate **211** (for example, a rear surface window) arranged on the second surface **2002**. According to an embodiment, a rear facing camera device **212** may be arranged on the rear surface plate **211**. At least one electronic component **213** may be arranged in the proximity of the rear facing camera device **212**. According to an embodiment, the electronic component **213** may include at least one of an illuminance sensor (for example, a light sensor), a proximity sensor (for example, a light sensor), an infrared sensor, an ultrasound sensor, a heartbeat sensor, a fingerprint recognition sensor, an EM sensor, or a flash device.

[0068] According to various embodiments, the display **201** may include a touch panel and a display panel which are layered on the rear surface of the front surface plate **2011**. According to an embodiment, an image displayed through the display panel may be provided to the user through the front surface plate **2011** of a transparent material. According to an embodiment, the front surface plate **2011** may use various materials such as transparent glass or acryl.

[0069] According to various embodiments, the electronic device **200** may include a waterproof structure. According to an embodiment, the electronic device **200** may include at least one waterproof member (sealing member) arranged therein to perform a waterproof function. According to an embodiment, the at least one waterproof member may be arranged between the display **201** and the side surface member **216** and/or between the side surface member **216** and the rear surface plate **211**.

[0070] According to various embodiments, the electronic device **200** may include at least one communication device (for example, a communication device **400** of FIG. 4A) using a millimeter wave (for example, a band of around 25 GHz or higher) as an operating frequency band. According to an embodiment, the communication device may include an antenna array including a plurality of antenna elements arranged on a dielectric (for example, a substrate) at regular intervals, and the antenna array may form a beam in at least one direction, and may transmit and receive signals in a beam forming direction through a communication circuit (for example, a radio frequency integrated circuit (RFIC) **311**, **321**, **331**, **341** of FIG. 3). According to an embodiment, a phase shifting means (for example, a phase shifter) (not shown) may be included in the proximity of the communication device.

[0071] According to various embodiments, the at least one communication device may be arranged in each corner of the electronic device. However, this should not be considered as limiting. The at least one communication device may be arranged in at least one portion of a rear surface and/or an

edge of the electronic device. According to an embodiment, even when a mounting position of the communication device is changed in the electronic device through appropriate arrangements of the antenna array and the communication circuit, optimal radiation performance can be implemented by considering only an arrangement direction without changing the design of the communication device.

[0072] FIG. 3 is a view illustrating an arrangement relationship of a communication device in an electronic device according to various embodiments of the present disclosure.

[0073] According to an embodiment, the electronic device **300** of FIG. 3 may be similar to the electronic device **101** of FIG. 1 or the electronic device **200** of FIG. 2A at least in part, or may include other embodiments of the electronic device.

[0074] Referring to FIG. 3, the electronic device **300** may include at least one communication device **310, 320, 330, 340**. According to an embodiment, the communication device **310, 320, 330, 340** may have antenna elements arranged on a substrate (printed circuit board (PCB)) at regular intervals in the form of at least one array, and may transmit and receive a signal for a designated direction through an RFIC **311, 321, 331, 341** arranged on the substrate. According to an embodiment, the antenna element may include a conductive member formed on the substrate in a patch type (or pattern type), or an antenna element of a dipole form

[0075] According to various embodiments, the electronic device **300** may include a PCB **350** (for example, a main PCB) mounted in an inner space thereof. According to an embodiment, the electronic device **300** may include a processor **370** (for example, a CP), an intermediate frequency IC **360** (for example, an IFIC), and a communication circuit which are mounted on the PCB **350**. According to an embodiment, the RFIC **311, 321, 331, 341** arranged in the communication device may be electrically connected to the intermediate frequency IC **360** through an electric connection member **381** (for example, a coaxial cable). According to an embodiment, a signal received through the communication device **310, 320, 330, 340** may be converted into an intermediate frequency signal through the RFIC **311, 321, 331, 341**, and the intermediate frequency signal may be changed to a base band frequency through the intermediate frequency IC **360**, and may be provided to the processor **370**.

[0076] According to various embodiments, the communication circuit **390** arranged on the PCB **350** may be electrically connected to a conductive member **391** (for example, the first portion **2101** of FIG. 2A) arranged in at least part of the electronic device, thereby transmitting and receiving a wireless signal through the conductive member **391**. According to an embodiment, the communication circuit **390** electrically connected with the conductive member **391** may provide wireless communication ranging from around 500 MHz to 6000 MHz. According to an embodiment, the RFIC **311, 321, 331, 341** included in the communication device **310, 320, 330, 340** and electrically connected with the plurality of antenna elements may provide wireless communication ranging from around 20 GHz to 100 GHz.

[0077] FIGS. 4A and 4B are perspective views illustrating a communication device according to various embodiments of the present disclosure.

[0078] According to an embodiment, the communication device **400** of FIG. 4A may be similar to the communication device **310, 320, 330, 340** of FIG. 3 at least in part, or may include other embodiments of the communication device.

[0079] Referring to FIGS. 4A and 4B, the communication device **400** may include a substrate **410** (for example, a dielectric). According to an embodiment, the substrate **410** may include a first surface **4001** and a second surface **4002** facing toward the opposite direction of the first surface **4001**. According to an embodiment, the communication device **400** may be arranged in an electronic device in such a manner that the first surface **4001** faces toward a rear surface plate (for example, the rear surface plate **211** of FIG. 2B) of the electronic device (for example, the electronic device **200** of FIG. 2B).

[0080] According to various embodiments, the substrate **410** may be formed in a substantially

rectangular shape. According to an embodiment, the substrate **410** may include a first side **411**, a second side **412** extending from the first side **411** perpendicular to the first side **411**, a third side **413** extending from the second side **412** perpendicular to the second side **412** and in parallel with the first side **411**, and a fourth side **414** extending from the third side **413** perpendicular to the third side **413** and in parallel with the second side **412**.

[0081] According to various embodiments, the communication device **400** may include at least one antenna array arranged on the first surface **4001** of the substrate **410**. According to an embodiment, the at least one antenna array may include a patch type or pattern type conductive member formed on the first surface **4001** of the substrate **410**. According to an embodiment, the antenna array may include a first antenna array **421** which forms a beam pattern to radiate toward the first side **411** of the substrate **410** (for example, the direction of {circle around (1)}), a second antenna array **422** which forms a beam pattern to radiate toward the second side **412** of the substrate **410** (for example, the direction of {circle around (2)}), and a third antenna array **423** which forms a beam pattern to radiate toward the rear surface plate (for example, the rear surface plate **211** of FIG. 2B) of the electronic device (for example, the direction of {circle around (5)}). According to an embodiment, the first antenna array **421** and the second antenna array **422** may include a dipole antenna radiator formed on the first surface **4001** of the substrate **410** in a pattern type. However, this should not be considered as limiting. The first antenna array **421** and the second antenna array **422** may be arranged on the second surface **4002** of the substrate **410** adjacent to the first side **411** and the second side **412** of the substrate **410**, respectively, or may be arranged on a side surface between the first surface **4001** and the second surface **4002**. According to an embodiment, the third antenna array **423** may include a patch type conductive member formed on the first surface **4001** of the substrate **410**. According to an embodiment, the communication device **400** may include only the third antenna array **423** and may not include the first antenna array **421** and the second antenna array **422** arranged on the first side **411** and the second side **412**, respectively, to radiate laterally.

[0082] According to various embodiments, the communication device **400** may include a communication circuit **430** (for example, an RFIC) arranged on the second surface **4002** of the substrate **410**. According to an embodiment, the antenna arrays **421**, **422**, **423** may be electrically connected with the communication circuit **430** through a conductive via penetrating from the first surface **4001** of the substrate **410** to the second surface **4002**. However, this should not be considered as limiting. The antenna arrays **421**, **422**, **423** may be fed through coupling with the communication circuit **430** (capacitively). According to an embodiment, the communication device **400** may include a shield can **440** arranged or mounted on the second surface **4002** of the substrate **410** to cover or surround at least a portion of the communication circuit **430**. According to an embodiment, the communication circuit **430** is shielded from a noise by the shield can **440**.

According to an embodiment, the communication device **400** may include at least one terminal **450**, **460** to be electrically connected with a PCB (for example, the PCB **350** of FIG. 3) of the electronic device (for example, the electronic device **300** of FIG. 3) through at least one portion of the substrate **410**. According to an embodiment, the terminal **450**, **460** may include a power terminal **450** and/or an RF terminal **460** to be electrically connected with the PCB of the electronic device through an electric connection member. According to an embodiment, the electric connection member may include a flexible PCB (FPCB) **451** or a coaxial cable **461**. According to an embodiment, it is illustrated that the electric connection member is divided into two or more members, but the electric connection members may be formed on a single FPCB altogether.

[0083] According to an embodiment, a drawing-out direction of the electric connection member may face toward the left or right of the communication circuit **430** or a lower end (for example, a lower end of the middle of the third antenna array **423**). According to an embodiment, the at least one terminal **450**, **460** may be arranged on a region of the substrate that is spaced apart from the beam pattern direction by the antenna array **421**, **422**, **423** among regions of the substrate **410**. According to an embodiment, the at least one terminal **450**, **460** may be arranged on a region of the

substrate **410** that is spaced apart from the beam pattern direction (for example, a radiation direction) (the direction of {circle around (1)}) of the first antenna array **421**, the beam pattern direction (for example, a radiation direction) (the direction of {circle around (2)}) of the second antenna array **422**, and the beam pattern direction (for example, a radiation direction) (the direction of {circle around (5)}) of the third antenna array **423**. According to an embodiment, the at least one terminal **450**, **460** may be arranged on a first part P1 that is a region of the third side **413** of the substrate **410** spaced apart from the first side **411** and the second side **412**, and/or a second part P2 that is a region of the fourth side **414** of the substrate **410** spaced apart from the first side **411** and the second side **412**. According to an embodiment, the at least one terminal **450**, **460** may be arranged on the third side **413** and/or the fourth side **414**, rather than on the first side **411** and the second side **412** on which the first antenna array **421** and the second antenna array **422** are arranged, spaced apart from the first and second antenna arrays **421**, **422** by a specified distance. [0084] FIG. 5 is a view illustrating an arrangement of a communication device according to various embodiments of the present disclosure. FIG. 5 illustrates a rear surface of an electronic device with a rear surface plate being removed when viewed from above.

[0085] According to an embodiment, the electronic device **500** of FIG. 5 may be similar to the electronic device **101** of FIG. 1, the electronic device **200** of FIG. 2A, or the electronic device of FIG. 3 at least in part, or may include other embodiments of the electronic device.

[0086] Referring to FIG. 5, the electronic device **500** may include a housing **510**. According to an embodiment, the housing **510** may include a side surface member **516**. According to an embodiment, the side surface member **516** may have at least one portion formed by a conductive member, and may be implemented as a unit conductive portion by a nonconductive portion, thereby operating as an antenna radiator. According to an embodiment, a first point **5105** of a first portion **5101** formed by a conductive member may be electrically connected with a communication circuit (for example, the communication circuit **390** of FIG. 3) of the electronic device **500**. According to an embodiment, the communication circuit may provide wireless communication ranging from around 500 MHz to 6000 MHz by using the first portion.

[0087] According to various embodiments, the housing **510** may include the first portion **5101** having a first length, a second portion **5102** extending perpendicular to the first portion **5101** and having a second length, a third portion **5103** extending from the second portion **5102** in parallel with the first portion **5101** to have the first length, and a fourth portion **5104** extending from the third portion **5103** in parallel with the second portion **5102** to have the second length.

[0088] According to various embodiments, at least one communication device **400**, **400-1**, **400-2**, **400-3** may be arranged in an inner space **5106** of the electronic device **500**.

[0089] According to an embodiment, the at least one communication device **400**, **400-1**, **400-2**, **400-3** may be arranged on a position which overlaps or does not overlap a PCB **520** (for example, a main PCB) of the electronic device. According to an embodiment, the PCB **520** may be arranged to avoid a battery **530**. The communication device **400**, **400-1**, **400-2**, **400-3** may be arranged to avoid overlapping the battery **530** and/or the PCB **520** or to overlap the battery **530** and/or the PCB **520** at least in part. According to an embodiment, the at least one communication device **400**, **400-1**, **400-2**, **400-3** may be arranged in at least one corner of the electronic device **500** which has a substantially rectangular shape. However, this should not be considered as limiting. The at least one communication device **400**, **400-1**, **400-2**, **400-3** may be arranged on an edge rather than in a corner or may be arranged in a corner and on an edge in combination. According to an embodiment, the first communication device **400** may be arranged to have a first side **411** be adjacent to the first portion **5101** of the housing **510**, and to have a second side **412** be adjacent to the second portion **5102** of the housing **510**. In this case, a second part P2 of the substrate **410** on which a power terminal (for example, the power terminal **450** of FIG. 4B) and an RF terminal (for example, the RF terminal **460** of FIG. 4B) are arranged may be arranged to face toward the center of the electronic device **500**, spaced apart from the beam pattern directions (for example, radiation

directions) (the directions of {circle around (1)} and {circle around (2)}) of the first communication device **400**. According to an embodiment, the second communication device **400-1** may be arranged in a corner between the second portion **5102** and the third portion **5103** of the housing **510** after being rotated by 90 degrees in the clockwise direction with reference to the first communication device **400**. In this case, the second part P2 of the second communication device **400-1** may also be arranged to face toward the center of the electronic device **500**, spaced apart from the beam pattern directions (for example, radiation directions) (the directions of {circle around (2)} and {circle around (3)}) of the second communication device **400-1**. According to an embodiment, the third communication device **400-2** may be arranged in a corner between the third portion **5103** and the fourth portion **5104** after being rotated by 90 degrees in the clockwise direction with reference to the second communication device **400-1**. In this case, the second part P2 of the third communication device **400-2** may also be arranged to face toward the center of the electronic device **500**, spaced apart from the beam pattern directions (for example, radiation directions) (the directions of {circle around (3)} and {circle around (4)}) of the third communication device **400-2**. According to an embodiment, the fourth communication device **400-3** may be arranged in a corner between the fourth portion **5104** and the first portion **5101** after being rotated by 90 degrees in the clockwise direction with reference to the third communication device **400-2**. In this case, the second part P2 of the fourth communication device **400-3** may also be arranged to face toward the center of the electronic device **500**, spaced apart from the beam pattern directions (for example, radiation directions) (the directions of {circle around (4)} and {circle around (1)}) of the fourth communication device **400-3**. According to an embodiment, the at least one communication device **400**, **400-1**, **400-2**, **400-3** may form a beam pattern in a direction toward the rear surface plate (for example, the direction of $-Z$ of FIG. 2A) by a third antenna array **423**.

[0090] According to various embodiments, the four communication devices **400**, **400-1**, **400-2**, **400-3** arranged in the electronic device **500** may have the same or similar configuration as or to that of the communication device of FIG. 4A and FIG. 4B. For example, the communication device of FIG. 4A and FIG. 4B may be applied to four corners of the electronic device by changing only its arrangement direction without separately changing a design. For example, the communication device **400**, **400-1**, **400-2**, **400-3** has the power terminal and the RF terminal implemented on a region of the substrate that is spaced apart from the beam pattern direction (for example, a direction facing toward the outside from the inside of the electronic device) of a wireless signal by the antenna array. Therefore, even when the arrangement position of the communication device, designed to have the same configuration, is changed by rotating the communication device, the power terminal and the RF terminal may always be arranged to face toward the center of the electronic device. Even when the arrangement position of the communication device according to an embodiment is changed, the design of each communication device may not be changed in consideration of the beam pattern direction.

[0091] FIGS. 6A and 6B are perspective views illustrating a communication device according to various embodiments of the present disclosure.

[0092] According to an embodiment, the communication device **600** of FIGS. 6A and 6B may be similar to the communication device **310**, **320**, **330**, **340** of FIG. 3 at least in part, or may include other embodiments of the communication device.

[0093] Referring to FIGS. 6A and 6B, unlike in FIGS. 4A and 4B, antenna elements **621**, **622** are arranged in the form of 1×4 in the communication device.

[0094] According to an embodiment, the communication device **600** may include a substrate **610** (for example, a dielectric). According to an embodiment, the substrate **610** may include a first surface **6001** and a second surface **6002** facing toward the opposite direction of the first surface **6001**. According to an embodiment, the communication device **600** may be arranged in an electronic device in such a manner that the first surface **6001** faces toward a rear surface plate (for

example, the rear surface plate **211** of FIG. 2B) of the electronic device (for example, the electronic device **200** of FIG. 2B).

[0095] According to various embodiments, the substrate **610** may be formed in a substantially rectangular shape. According to an embodiment, the substrate **610** may include a first side **611**, a second side **612** extending from the first side **611** perpendicular to the first side **611**, a third side **613** extending from the second side **612** perpendicular to the second side **612** and in parallel with the first side **611**, and a fourth side **614** extending from the third side **613** perpendicular to the third side **613** and in parallel with the second side **612**.

[0096] According to various embodiments, the communication device **600** may include at least one antenna array (for example, a first antenna array **621**, or a second antenna array **622**) arranged on the first surface **6001** of the substrate **610**. According to an embodiment, the plurality of antenna arrays **621**, **622** may include a patch type or pattern type conductive member formed on the first surface **6001** of the substrate **610**. According to an embodiment, the antenna arrays may include the first antenna array **621** which forms a beam pattern to radiate toward the first side **611** (for example, the direction of {circle around (1)}) of the substrate **610**, and the second antenna array **622** which forms a beam pattern to radiate toward the rear surface plate (for example, the rear surface plate **211** of FIG. 2B) (for example, the direction of {circle around (5)}) of the electronic device. According to an embodiment, the first antenna array **621** may include a dipole antenna radiator formed on the first surface **6001** of the substrate **610** in a pattern type. However, this should not be considered as limiting. The first antenna array **621** may be arranged on the second surface **6002** of the substrate **610** adjacent to the first side **611** of the substrate **610**, or may be arranged on a side surface between the first surface **6001** and the second surface **6002**. According to an embodiment, the second antenna array **622** may include a patch type conductive member formed on the first surface **6001** of the substrate **610**.

[0097] According to various embodiments, the communication device **600** may include a communication circuit **630** (for example, an RFIC) arranged on the second surface **6002** of the substrate **610**. According to an embodiment, the antenna arrays **621**, **622** may be electrically connected with the communication circuit **630** through a conductive via penetrating from the first surface **6001** of substrate **610** to the second surface **6002**. However, this should not be considered as limiting. The antenna arrays **621**, **622** may be fed through coupling (capacitively) of at least portions of a feeding line with the communication circuit **630**. According to an embodiment, the communication device **600** may include a shield can **640** arranged or mounted on the second surface **6002** of the substrate **610** to cover or surround at least a portion of the communication circuit **630**. According to an embodiment, the communication circuit **630** is shielded from a noise by the shield can **640**. According to an embodiment, the communication device **600** may include at least one terminal **650**, **660** to be electrically connected with a PCB (for example, the PCB **350** of FIG. 3) of an electronic device (for example, the electronic device **300**) through at least one portion of the substrate **610**. According to an embodiment, the terminal **650**, **660** may include a power terminal **650** and an RF terminal **660** to be electrically connected with the PCB of the electronic device through an electric connection member. According to an embodiment, the electric connection member may include an FPCB **651** or a coaxial cable **661**. According to an embodiment, it is illustrated that the electric connection member is divided into two or more members, but the electric connection members may be configured on a single FPCB altogether. According to an embodiment, a drawing-out direction of the electric connection member may face toward the left or right of the communication circuit **630** or a lower end (a lower end of the middle of the second antenna array) of the communication circuit **630**. According to an embodiment, the at least one terminal **650**, **660** may be arranged on a region of the substrate that is spaced apart from the beam pattern direction by the antenna array **621**, **622** among regions of the substrate **610**. According to an embodiment, the at least one terminal **650**, **660** may be arranged on a region of the substrate **610** that is spaced apart from the beam pattern direction (for example, a radiation

direction) (the direction of {circle around (1)}) of the first antenna array **621**, and the beam pattern direction (for example, a radiation direction) (the direction of {circle around (5)}) of the second antenna array **622**. According to an embodiment, the at least one terminal **650**, **660** may be arranged on any one of the other sides **612**, **613**, **614** than the first side **611** on which the first antenna array **621** is arranged, spaced apart from the first antenna array **621** by a specified distance. [0098] FIGS. 7A and 7B are views illustrating an arrangement of a communication device according to various embodiments of the present disclosure. FIGS. 7A and 7B are views illustrating a configuration in which the communication circuit of FIG. 6 is arranged in an electronic device. [0099] According to an embodiment, the electronic device **700** of FIGS. 7A and 7B may be similar to the electronic device **200** of FIG. 2A at least in part, or may include other embodiments of the electronic device. In explaining the drawings, a configuration in which the communication device is electrically connected with a PCB arranged in the electronic device is similar to the above-described configuration, and thus a detailed description thereof is omitted, and an arrangement structure of the communication device will be mainly described.

[0100] Referring to FIG. 7A, the electronic device **700** may include a housing **710**. According to an embodiment, the housing **710** may include a side surface member **716**. According to an embodiment, at least one portion of the side surface member **716** may be formed by a conductive member, and is implemented as a unit conductive portion by a nonconductive portion, thereby operating as an antenna radiator.

[0101] According to various embodiments, the housing **710** may have a first portion **7101** having a first length, a second portion **7102** extending perpendicular to the first portion **7101** and having a second length, a third portion **7103** extending from the second portion **7102** in parallel with the first portion **7101** to have the first length, and a fourth portion **7104** extending from the third portion **7103** in parallel with the second portion **7102** to have the second length.

[0102] According to various embodiments, at least one communication device **600**, **600-1** may be arranged in an inner space **7106** of the electronic device **700**. According to an embodiment, the at least one communication device **600**, **600-1** may be arranged in at least one corner of the electronic device **700** having a substantially rectangular shape.

[0103] According to various embodiments, a first side **611** of the first communication device **600** may be arranged adjacent to the first portion **7101** of the housing **710**, and a second side **612** of the first communication device **600** may be arranged adjacent to the second portion **7102** of the housing **710**. In this case, a power terminal (for example, the power terminal **650** of FIG. 6B) and an RF terminal (for example, the RF terminal **660** of FIG. 6B) may be drawn out from a fourth side **614** toward the center of the electronic device **700**. In another example, the power terminal and the RF terminal may be drawn out from a third side **613** toward the center of the electronic device. According to an embodiment, a first side **611** of the second communication device **600-1** may be arranged adjacent to the fourth portion **7104** of the housing **710**, and a second side **612** of the second communication device **600-1** may be arranged adjacent to the first portion **7101** of the housing **710**.

[0104] According to various embodiments, the first communication device **600** may form a beam pattern facing toward the first portion **7101** (for example, toward the direction of {circle around (1)}) of the housing **710** by a first antenna array **621**, and may form a beam pattern facing toward a rear surface plate (for example, in the direction of $-Z$ of FIG. 2A) of the electronic device by a second antenna array **622**. According to an embodiment, the second communication device **600-1** may form a beam pattern facing toward the fourth portion **7104** (for example, toward the direction of {circle around (4)}) of the housing **710** by a first antenna array **621**, and may form a beam pattern facing toward the rear surface plate (for example, in the direction of $-Z$ of FIG. 2A) of the electronic device by a second antenna array **622**.

[0105] Referring to FIG. 7B, at least one communication device **600**, **600-1**, **600-2** may be arranged on a certain region of each edge of the electronic device **700**. According to an

embodiment, a first side **611** of the first communication device **600** may be arranged adjacent to and in parallel with the first portion **7101** on a substantially center of the first portion **7101** of the housing **710**. According to an embodiment, a first side **611** of the second communication device **600-1** may be arranged adjacent to and in parallel with the fourth portion **7104** on a certain region of the fourth portion **7104** of the housing **710**. According to an embodiment, a first side **611** of the third communication device **600-2** may be arranged adjacent to and in parallel with the second portion **7102** on a certain region of the second portion **7102** of the housing **710**.

[0106] According to various embodiments, the first communication device **600** may form a beam pattern facing toward the first portion **7101** (for example, toward the direction of {circle around (1)}) of the housing **710** by a first antenna array **621**, and may form a beam pattern facing toward the rear surface plate (for example, in the direction of $-Z$ of FIG. 2A) of the electronic device by a second antenna array **622**. According to an embodiment, the second communication device **600-1** may form a beam pattern facing toward the fourth portion **7104** (for example, toward the direction of {circle around (4)}) of the housing **710** by a first antenna array **621**, and may form a beam pattern facing toward the rear surface plate (for example, in the direction of $-Z$ of FIG. 2A) of the electronic device by a second antenna array **622**. According to an embodiment, the third communication device **600-2** may form a beam pattern facing toward the second portion **7102** (for example, toward the direction of {circle around (2)}) of the housing **710** by a first antenna array **621**, and may form a beam pattern facing toward the rear surface plate (for example, in the direction of $-Z$ of FIG. 2A) of the electronic device by a second antenna array **622**.

[0107] According to various embodiments, a region of the housing **710** corresponding to a portion on which the communication device **600**, **600-1**, **600-2** is mounted may be formed of a material (for example, a dielectric substance) other than a conductive material to prevent degradation of radiation performance of the communication device. However, this should not be considered as limiting. The corresponding region of the housing may be substituted with a hole formed on the housing in the beam forming direction of the communication device, or with a metal-organic framework (for example, a metal grid) through which beams can pass.

[0108] FIGS. **8A**, **8B**, and **8C** are views illustrating an arrangement of a communication device according to various embodiments of the present disclosure.

[0109] FIGS. **8A**, **8B**, and **8C** are views illustrating a configuration in which the communication circuit of FIG. **6** is arranged in an electronic device.

[0110] The electronic device **800** of FIGS. **8A**, **8B**, and **8C** may be similar to the electronic device **200** of FIG. 2A at least in part, or may include other embodiments of the electronic device. In explaining the drawings, a configuration in which a communication device **600**, **600-1** is electrically connected with a PCB arranged in the electronic device **800** is similar to the above-described configuration, and thus a detailed description thereof is omitted, and an arrangement structure of the communication device **600**, **600-1** will be mainly described.

[0111] Referring to FIGS. **8A** and **8B**, the electronic device **800** may include a housing **810**. According to an embodiment, the housing **810** may include a side surface member **816**. According to an embodiment, at least one portion of the side surface member **816** may be formed by a conductive member, and may be implemented as a unit conductive portion by a nonconductive portion, thereby operating as an antenna radiator.

[0112] According to various embodiments, the housing **810** may include a first portion **8101** having a first length, a second portion **8102** extending perpendicular to the first portion **8101** and having a second length, a third portion **8103** extending from the second portion **8102** in parallel with the first portion **8101** to have the first length, and a fourth portion **8104** extending from the third portion **8103** in parallel with the second portion **8102** to have the second length.

[0113] According to various embodiments, at least one communication device **600**, **600-1** may be arranged in an inner space **8106** of the electronic device **800**. According to an embodiment, the at least one communication device **600**, **600-1** may be arranged adjacent to at least one side of the

electronic device **800** having a substantially rectangular shape. According to an embodiment, the at least one communication device **600**, **600-1** may be arranged in such a manner that a first surface **6001** faces toward the side surface member **816** of the housing **810**.

[0114] According to various embodiments, the first communication device **600** may be arranged on a position adjacent to the first portion **8101** among regions where the first portion **8101** and the second portion **8102** of the housing **810** meet in the inner space **8106** of the electronic device **800** to have the first surface **6001** face toward the first portion **8101**. According to an embodiment, the second communication device **600-1** may be arranged on a position adjacent to the fourth portion **8104** among regions where the first portion **8101** and the fourth portion **8104** of the housing **810** meet to have a first surface **6001** face toward the fourth portion **8104**.

[0115] According to various embodiments, the first communication device **600** may form a beam pattern facing toward the rear surface plate of the electronic device (for example, in the direction of $-Z$ of FIG. 2A) (for example, the direction of {circle around (5)}) by a first antenna array **621**, and may form a beam pattern facing toward the first portion **8101** of the housing **810** (for example, toward the direction of {circle around (1)}) by a second antenna array **622**. According to an embodiment, the second communication device **600-1** may form a beam pattern facing toward the rear surface plate of the electronic device (for example, in the direction of $-Z$ of FIG. 2A) (for example, the direction of {circle around (5)}) by a first antenna array **621**, and may form a beam pattern facing toward the fourth portion **8104** of the housing **810** (for example, toward the direction of {circle around (4)}) by a second antenna array **622**.

[0116] Referring to FIG. 8C, at least one communication device **600**, **600-1**, **600-2** may be arranged on a certain region of an edge of the electronic device **800**. According to an embodiment, the first communication device **600** may be arranged to have the first surface **6001** face the first portion **8101** on a substantially center of the first portion **8101** of the housing **810**. According to an embodiment, the second communication device **600-1** may be arranged to have the first surface **6001** be adjacent to and in parallel with the fourth portion **8104** on a certain region of the fourth portion **8104** of the housing **810**. According to an embodiment, the third communication device **600-2** may be arranged to have the first surface **6001** be adjacent to and in parallel with the second portion **8102** on a certain region of the second portion **8102** of the housing **810**.

[0117] According to various embodiments, the first communication device **600** may form a beam pattern facing toward the rear surface plate of the electronic device (for example, in the direction of $-Z$ of FIG. 2A) (for example, the direction of {circle around (5)}) by a first antenna array **621**, and may form a beam pattern facing toward the first portion **8101** of the housing **810** (for example, toward the direction of {circle around (1)}) by a second antenna array **622**. According to an embodiment, the second communication device **600-1** may form a beam pattern facing toward the rear surface plate of the electronic device (for example, in the direction of $-Z$ of FIG. 2A) (for example, the direction of {circle around (5)} of FIG. 8A) by a first antenna array **621**, and may form a beam pattern facing toward the fourth portion **8104** of the housing **810** (for example, toward the direction of {circle around (4)}) by a second antenna array **622**. According to an embodiment, the third communication device **600-2** may form a beam pattern facing toward the rear surface plate of the electronic device (for example, in the direction of $-Z$ of FIG. 2A) (for example, the direction of {circle around (5)} of FIG. 8A) by a first antenna array **621**, and may form a beam pattern facing toward the second portion **8102** of the housing **810** (for example, toward the direction of {circle around (2)}) by a second antenna array **622**.

[0118] According to various embodiments, although not shown, the communication device **600** illustrated in FIGS. 6A and 6B may be arranged in at least one portion of each corner or each edge of the electronic device having the substantially rectangular shape, or may be arranged in a corner or on an edge in combination.

[0119] According to various embodiments, a region of the housing **810** corresponding to a portion on which the communication device **600**, **600-1**, **600-2** is mounted may be formed of a material

(for example, a dielectric substance) other than a conductive material to prevent degradation of radiation performance of the communication device. However, this should not be considered as limiting. The corresponding region of the housing may be substituted with a hole formed on the housing in the beam forming direction of the communication device, or with a metal-organic framework (for example, a metal grid) through which beams can pass.

[0120] FIG. 9 is a view illustrating an arrangement structure of communication devices of an electronic device according to various embodiments of the present disclosure.

[0121] According to an embodiment, the electronic device **900** of FIG. 9 may be similar to the electronic device **101** of FIG. 1, the electronic device **200** of FIG. 2A, the electronic device **300** of FIG. 3, the electronic device **500** of FIG. 5, the electronic device **700** of FIGS. 7A and 7B, or the electronic device **800** of FIGS. 8A, 8B, and 8C at least in part, or may include other embodiments of the electronic device.

[0122] Referring to FIG. 9, the electronic device **900** may include a housing **910**. According to an embodiment, the housing **910** may be formed of a conductive member and/or a nonconductive member. According to an embodiment, the housing **910** may include a first conductive portion **911** having a first length, a second conductive portion **912** extending perpendicular to the first conductive portion **911** and having a second length, and a third conductive portion **913** extending perpendicular to the first conductive portion **911** and in parallel with the second conductive portion **912**. Although not shown, the housing **910** may include a fourth conductive portion (for example, the third portion **5103** of FIG. 5) extending from an end of the second conductive portion **912** to the third conductive portion **913** and arranged in parallel with the first conductive portion **911**. According to an embodiment, a side surface member **916** may include unit conductive portions **911**, **912**, **913** which are electrically isolated from one another by a pair of nonconductive portions **9101**, **9102** spaced apart from each other by a predetermined distance on the first conductive portion **911**.

[0123] According to various embodiments, at least one portion of the first conductive portion **911** may be electrically connected with a communication circuit (for example, the communication circuit **390** of FIG. 3) mounted on a PCB (for example, the PCB **350** of FIG. 3) of the electronic device **900** at a first point **9111**, thereby operating as a first antenna **A1**. According to an embodiment, at least one portion of the second conductive portion **912** may be electrically connected with the communication circuit mounted on the PCB of the electronic device at a second point **9121**, thereby operating as a second antenna **A2**. At least one portion of the third conductive portion **913** may be electrically connected with the communication circuit mounted on the PCB of the electronic device at a third point **9131** and a fourth point **9132**, thereby operating as a third antenna **A3** and a fourth antenna **A4**. According to an embodiment, each point may be physically connected to the PCB by using a C-clip, a conductive contact, or the like, or may be electrically connected to the PCB by an electric connection member such as an FPCB. According to an embodiment, the first antenna **A1** may contribute as a multi-band antenna operating in a low band and a mid band. According to an embodiment, the second antenna **A2** may contribute as a short-range wireless communication antenna operating in a WiFi band and a BT band. According to an embodiment, the third antenna **A3** may contribute as an antenna operating in a high band, a mid band, and as a GPS. According to an embodiment, the fourth antenna **A4** may contribute as an antenna operating in a high band.

[0124] According to various embodiments, the electronic device **900** may include a first communication device **400** arranged in a corner between the first conductive portion **911** and the second conductive portion **912**, or a second communication device **400-1** arranged in a corner between the first conductive portion **911** and the third conductive portion **913**. According to an embodiment, the first communication device **400** or the second communication device **400-1** may be the same or similar module, and may perform wireless communication in a band ranging from around 20 GHz to 100 GHz. However, this should not be considered as limiting, and the first

communication device **400** and the second communication device **400-1** may be substituted with the communication device **600** of FIG. 6A. According to an embodiment, the first communication device **400** may form a beam pattern toward the first conductive portion (the direction of {circle around (1)}) and the second conductive portion (the direction of {circle around (2)}) as shown in FIG. 9. The second communication device **400-1** may form a beam pattern toward the first conductive portion (the direction of {circle around (1)}) and the third conductive portion (the direction of {circle around (4)})

[0125] According to various embodiments, at least part of the first conductive portion **911**, the second conductive portion **912**, and the third conductive portion **913** of the housing **910** arranged adjacent to the first and second communication devices **400**, **400-1** may influence the beam patterns of the first and second communication devices **400**, **400-1**. For example, as shown in FIG. 11A, the beam pattern of the communication device **400** may be tilted in an unintended direction (for example, tilted by 0) by a corresponding region (for example, the first conductive portion) of the housing **1110** overlapping the beam pattern direction of the communication device **400** arranged in the electronic device, and tilting of the beam pattern may induce degradation of radiation efficiency of the communication device **400**. According to an embodiment, the region overlapping the beam pattern direction of the communication device **400** may be formed to be void of a conductive portion.

[0126] FIGS. 10A, 10B, and 10C are views illustrating a configuration of a housing of an electronic device according to various embodiments of the present disclosure.

[0127] Referring to FIGS. 10A, 10B, and 10C, at least part of the housing **910** may be formed of a conductive member. According to an embodiment, the communication device **400** may be arranged on at least one portion of the housing **910**, and at least part of the conductive member of the first conductive portion **911** that overlaps the beam pattern direction (the direction of {circle around (1)}) of the communication device **400** may be excluded. According to an embodiment, at least part of the conductive members of the second conductive portion **912** and the third conductive portion **913** may be formed to exclude a region overlapping the beam pattern direction of the communication device **400** and/or **400-1**. According to an embodiment, the excluded region of the housing **910** described above may be formed on the housing by using a nonconductive member **940** through double injection molding when necessary. For example, when the corresponding region of the first conductive portion **911** overlapping the beam pattern direction of the communication device **400** is excluded as shown in FIG. 11B, directivity of the beam pattern is constantly maintained in a specified direction, and this can be interpreted as indicating that radiation performance of the communication device **400** and/or **400-1** is constantly maintained.

[0128] Although not shown, in addition to the conductive portions of the housing arranged in the proximity of the communication device, conductive electronic components (for example, a speaker module, a microphone module, an interface connector port, an ear jack assembly, a sensor module, or various kinds of switches) arranged in the electronic device may be arranged to avoid overlapping the beam pattern direction of the communication device, or at least one portion thereof may be substituted with a nonconductive member, such that degradation of radiation performance of the communication device can be prevented.

[0129] According to various embodiments, an electronic device (for example, the electronic device **200**) may include: a housing (for example, the housing **210** of FIG. 2A) including: a front surface plate (the front surface plate **2011** of FIG. 2A); a rear surface plate (the rear surface plate **211** of FIG. 2A) facing toward the opposite direction of the front surface plate; and a side surface member (for example, the side surface member **216** of FIG. 2A) surrounding a space between the front surface plate and the rear surface plate, the side surface member having a substantially rectangular shape when viewed above the front surface plate, the side surface member including: a first portion (for example, the first portion **2101** of FIG. 2A) having a first length and extending in a first direction; a second portion (for example, the second portion **2102** of FIG. 2A) having a second

length longer than the first length, and extending in a second direction perpendicular to the first direction; a third portion (for example, the third portion **2103** of FIG. 2A) having the first length and extending in the first direction in parallel with the first portion; and a fourth portion (for example, the fourth portion **2104** of FIG. 2A) having the second length and extending in the second direction in parallel with the second portion; a first PCB (for example, the PCB **520** of FIG. 5) arranged in the space; a first wireless communication circuit (for example, the communication circuit **390** of FIG. 3) which is mounted on the first PCB and is electrically connected with a first point (for example, the first point **5105** of FIG. 5) of the first portion of the side surface member, and provides wireless communication ranging from around 500 MHz to 6000 MHz; a substrate (for example, the substrate **410** of FIG. 5) arranged close to a corner in the space in parallel with the rear surface plate, the substrate including: a first side (for example, the first side **411** of FIG. 5) extending along a part of the first portion, a second side (for example, the second side **412** of FIG. 5) extending along the second portion; a third side (for example, the third side **413** of FIG. 5) extending in parallel with the first side; and a fourth side (for example, the fourth side **414** of FIG. 5) extending in parallel with the second side; a first antenna array (for example, the first antenna array **421** of FIG. 5) protruding from the first side of the substrate toward the first portion; a second antenna array (for example, the second antenna array **422** of FIG. 5) protruding from the second side of the substrate toward the second portion; and a second wireless communication circuit (for example, the RFIC **311, 321, 331, 341** of FIG. 3) attached to a first part (for example, the first part **P1** of FIG. 4A and FIG. 4B) on the third side closer to the fourth side than the second portion, or attached to a second part (for example, the second part **P2** of FIG. 4A and FIG. 4B) on the fourth side closer to the third side than the first portion, and the second wireless communication circuit may be electrically connected to the first antenna array and the second antenna array, and provides wireless communication ranging from around 20 GHz to 100 GHz.

[0130] According to various embodiments, the electronic device may further include an FPCB attached to the first part or the second part, and the second wireless communication circuit may be mounted on the FPCB.

[0131] According to various embodiments, the electronic device may further include a third antenna array arranged on the substrate and facing toward the rear surface plate.

[0132] According to various embodiments, the second wireless communication circuit may include an RF circuit and a power connector.

[0133] According to various embodiments, the first portion may be divided into unit conductive portions by a pair of nonconductive portions having the first point therebetween.

[0134] According to various embodiments, the substrate may further include: a first surface including the first antenna array and the second antenna array; and a second surface opposite the first surface and having the second wireless communication circuit mounted thereon.

[0135] According to various embodiments, the substrate may be arranged in the space to have the first surface face toward the rear surface plate.

[0136] According to various embodiments, the substrate may be arranged adjacent to a corner at which the first portion and the second portion meet.

[0137] According to various embodiments, the electronic device may further include at least one conductive member arranged in a direction in which the first antenna array protrudes toward the first portion, and/or a direction in which the second antenna array protrudes toward the second portion, and a region of the conductive member that overlaps a portion protruding toward the first portion, and/or a region of the conductive member that overlaps a portion protruding toward the second portion may exclude a conductive portion.

[0138] According to various embodiments, the excluded region may be substituted with a nonconductive member.

[0139] According to various embodiments, the conductive member may include at least one portion of the first portion and/or the second portion.

[0140] According to various embodiments, the conductive member may include at least one of a speaker module, a microphone module, an interface connector port, an ear jack assembly, a sensor module, or various kinds of switches arranged in the electronic device.

[0141] According to various embodiments, an electronic device may include: a housing including: a front surface plate; a rear surface plate facing toward the opposite direction of the front surface plate; and a side surface member surrounding a space between the front surface plate and the rear surface plate, the side surface member having a substantially rectangular shape when viewed above the front surface plate, the side surface member including: a first portion having a first length and extending in a first direction; a second portion having a second length longer than the first length and extending in a second direction perpendicular to the first direction; a third portion having the first length and extending in the first direction in parallel with the first portion; and a fourth portion having the second length and extending in the second direction in parallel with the second portion; a first substrate arranged in the space; a first wireless communication circuit which is mounted on the first substrate and is electrically connected with a first point of the first portion of the side surface member, and provides wireless communication of a first frequency band; a second substrate arranged in the proximity of a corner between the first portion and the second portion in the space in parallel with the rear surface plate, the second substrate including: a first side extending along a part of the first portion, a second side extending along the second portion; a third side extending in parallel with the first side; and a fourth side extending in parallel with the second side; a first antenna array arranged to form a beam pattern in a direction from the first side of the second substrate toward the first portion; a second antenna array arranged to form a beam pattern in a direction from the second side of the second substrate toward the second portion; a third substrate having a same configuration as that of the second substrate, and arranged in the proximity of a corner between the second portion and the third portion in the space in parallel with the rear surface plate, after being rotated by 90 degrees in the clockwise direction with reference to the arrangement state of the second substrate; a fourth substrate having a same configuration as that of the second substrate, and arranged in the proximity of a corner between the third portion and the fourth portion in the space in parallel with the rear surface plate, after being rotated by 90 degrees in the clockwise direction with reference to the arrangement state of the third substrate; a fifth substrate having a same configuration as that of the second substrate, and arranged in the proximity of a corner between the fourth portion and the first portion in the space in parallel with the rear surface plate, after being rotated by 90 degrees in the clockwise direction with reference to the arrangement state of the fourth substrate; and a second wireless communication circuit attached to a first part on the third side closer to the fourth side than the second portion, or attached to a second part on the fourth side closer to the third side than the first portion, and the second wireless communication circuit may be electrically connected to the first antenna array and the second antenna array and provides wireless communication in a second frequency band.

[0142] According to various embodiments, the first frequency band may include a frequency band ranging from around 500 MHz to 6000 MHz.

[0143] According to various embodiments, the second frequency band may include a frequency band ranging from around 20 GHz to 100 GHz.

[0144] According to various embodiments, the electronic device may further include at least one conductive member arranged in a beam pattern direction of the first antenna array and/or in a beam pattern direction of the second antenna array, and a region of the conductive member that overlaps the beam pattern direction of the first antenna array and/or the second antenna array may exclude a conductive portion.

[0145] According to various embodiments, the excluded region may be substituted with a nonconductive member.

[0146] According to various embodiments, the conductive member may include at least one portion of at least one of the first portion, the second portion, the third portion, and/or the fourth portion.

[0147] According to various embodiments, the conductive member may include at least one of a speaker module, a microphone module, an interface connector port, an ear jack assembly, a sensor module, or various kinds of switches arranged in the electronic device.

[0148] According to various embodiments, the second wireless communication circuit may include an RF circuit and a power connector.

[0149] The present disclosure has been described with reference to various example embodiments thereof. It will be understood by a person skilled in the art that the present disclosure can be implemented in modified forms without departing from the essential characteristics of the present disclosure. Therefore, disclosed embodiments should be considered from a descriptive perspective, not from a limited perspective. The scope of the present disclosure is defined not by the detailed description but by the appended claims, and all differences within the scope should be understood as being included in the present disclosure.

[0150] Although the present disclosure has been described with various embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims.

Claims

1. A portable communication device comprising: a housing including a conductive side surface member substantially surrounding a space between a front surface and a rear surface of the portable communication device, wherein the conductive side surface member includes a hole portion, and wherein a dielectric is disposed in the hole portion; and an antenna module including a substrate and an antenna array, wherein the antenna array includes a first antenna and a second antenna and is formed in the substrate, and wherein the antenna module is accommodated in the housing such that a first surface of the substrate is substantially parallel with the conductive side surface member, and that the first antenna and the second antenna face the hole portion.
2. The portable communication device of claim 1, wherein each of the first antenna and the second antenna is formed as a patch antenna.
3. The portable communication device of claim 1, further comprising: a radio frequency integrated circuit (RFIC) configured to transmit, to the antenna array, a radio frequency (RF) signal having a frequency band ranging from 20 GHz to 100 GHz.
4. The portable communication device of claim 3, wherein the RFIC is arranged on the substrate and faces away from the hole portion.
5. The portable communication device of claim 4, wherein the substrate includes a second surface opposite the first surface of the substrate, and wherein the RFIC is arranged on the second surface of the substrate.
6. The portable communication device of claim 5, wherein the RFIC is electrically connected to the antenna array through a conductive via penetrating at least a portion of the substrate.
7. The portable communication device of claim 1, further comprising: an intermediate frequency integrated circuit (IFIC) disposed on a substrate, wherein the IFIC is electrically connected to a radio frequency integrated circuit (RFIC) of the antenna module via a flexible printed circuit board (FPCB).
8. The portable communication device of claim 1, further comprising: a metal grid through which beams can pass, disposed in the hole portion.
9. The portable communication device of claim 1, wherein the antenna array is configured to radiate signals in a form of a beam that can pass through the hole portion.
10. The portable communication device of claim 1, wherein the dielectric within the hole portion includes an exterior surface to face an outside of the portable communication device, and an interior surface opposite to the exterior surface, and wherein each of the first antenna and the

second antenna faces the interior surface of the dielectric.

11. A portable communication device comprising: a housing including a conductive side surface member substantially surrounding a space between a front surface and a rear surface of the portable communication device, wherein the conductive side surface member includes a hole portion, and wherein a dielectric is disposed in the hole portion; and an antenna module including a substrate and an antenna array, wherein the antenna array includes a first antenna and a second antenna and is formed in the substrate, wherein the antenna module is accommodated in the housing such that a first surface of the substrate is substantially parallel with the conductive side surface member, and that the first antenna and the second antenna face the hole portion, and wherein each of the first antenna and the second antenna includes a patch antenna.

12. The portable communication device of claim 11, wherein the dielectric is formed by an injection molding.

13. The portable communication device of claim 11, further comprising: a radio frequency integrated circuit (RFIC) configured to transmit, to the antenna array, a radio frequency (RF) signal having a frequency band ranging from 20 GHz to 100 GHz.

14. The portable communication device of claim 13, wherein the RFIC is arranged on the substrate and faces away from the hole portion.

15. The portable communication device of claim 14, wherein the substrate includes a second surface opposite the first surface of the substrate, and wherein the RFIC is arranged on the second surface of the substrate.

16. The portable communication device of claim 15, wherein the RFIC is electrically connected to the antenna array through a conductive via penetrating at least a portion of the substrate.

17. The portable communication device of claim 11, further comprising: an intermediate frequency integrated circuit (IFIC) disposed on a substrate, wherein the IFIC is electrically connected to a radio frequency integrated circuit (RFIC) of the antenna module via a flexible printed circuit board (FPCB).

18. The portable communication device of claim 11, further comprising: a metal grid through which beams can pass, disposed in the hole portion.

19. The portable communication device of claim 11, wherein the antenna array is configured to radiate signals in a form of a beam that can pass through the hole portion.

20. The portable communication device of claim 11, wherein the dielectric within the hole portion includes an exterior surface to face an outside of the portable communication device, and an interior surface opposite to the exterior surface, and wherein each of the first antenna and the second antenna faces the interior surface of the dielectric.
